

By: AKH
 Rev.--- By: ---
 Chk: ---

Date May 2026
 Date ---
 Date ---

Macaulay Test

Macaulay Beam Analysis - BetaTesting

04 - Steel - 3 Point Loads at Quarter Spans

Bridge, L = in

Beam Sect = W14x61

A = 17.90 in²I = 640.00 in⁴E = 29,000,000 lb / in²G = 1.740E+06 lb / in² = 0.06*29,000,000

Length Units in

Force Units lb

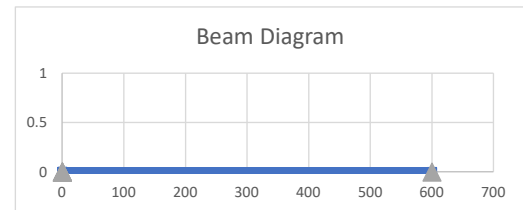
Beam Segments (First Segment Begins at 0)

	Segment No.	X End	A	I	E	G
		in	in ²	in ⁴	lb / in ²	lb / in ²
DO NOT DELETE ROWS (GRAPHED)	1	600.00	17.90	640.00	2.900E+07	3.115E+07
	2					
	3					
	4					
	5					

Supports

Perscribed Disp. (typ 0)

	Supp. No.	X	dY (vert)	dr (rot)
		in	in	rad
DO NOT DELETE ROWS (GRAPHED)	1	0.0	0.0	0.000
	2	600.0	0.0	0.000
	3			
	4			
	5			



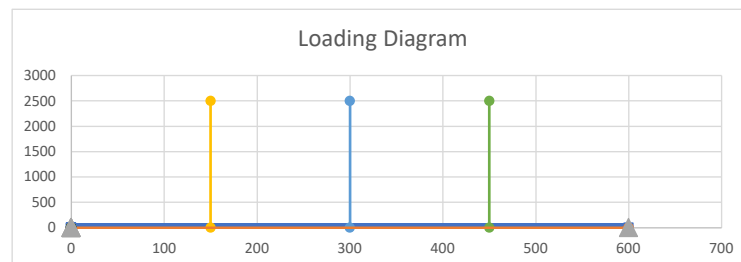
Distributed Loads (downward loads entered as NEGATIVE) - UNFACTORED

	Load No.	x1	x2	w1 (at x1)	w2 (at x2)
		in	in	lb / in	lb / in
DO NOT DELETE ROWS (GRAPHED)	1	0.0	600.0	0.0	0.0
	2	0.0	0.0	0.0	0.0
	3	0.0	0.0	0.0	0.0
	4	0.0	0.0	0.0	0.0
	5	0.0	0.0	0.0	0.0

Load
lb / ft
0.0
0.0
0.0
0.0

Point Loads - UNFACTORED

	Load No.	X	Py	My
		in	lb	lb-in
DO NOT DELETE ROWS (GRAPHED)	1	150	-2500.00	0.00
	2	300	-2500.00	0.00
	3	450	-2500	0.00
	4			
	5			



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BEAM REACTIONS

Position (X)	Enforced Disp (dy)	Enforced Rot (theta)	Vert Reac (Ry)	Internal Moment (Mz)	Note
(L)	(L)	(rad)	(F)	(F-L)	(---)
0.00	0.0	0.0	3,750.0	0.0	Support reaction
600.00	0.00	0.00	3,750.0	0.0	Support reaction

Vert Reac
(kip)
3.750
3.750

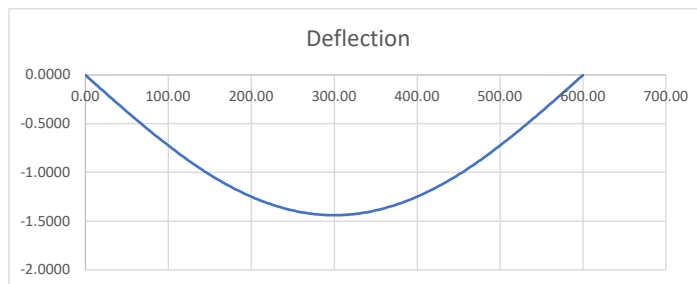
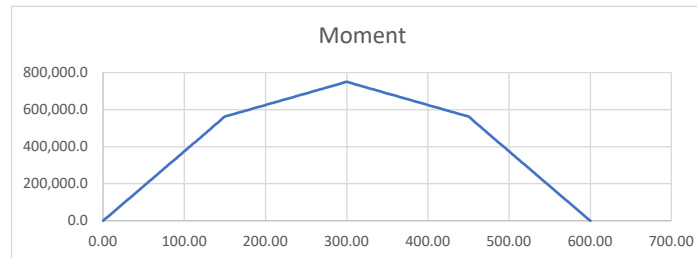
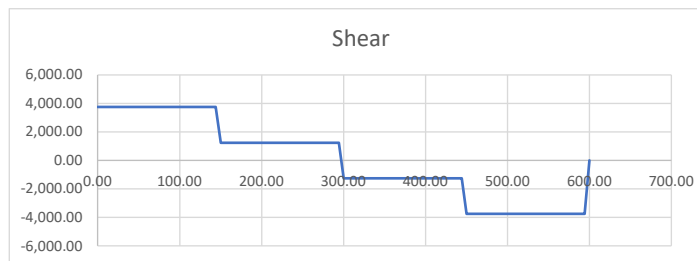
BEAM REACTIONS EQUILIBRIUM CHECK

Item	Units	Loads	Reactions	Residual	Check Sum
Sum Fy	F	-7,500.0	7,500.0	0.0	OK
Sum M @ x=0	F-L	-2.2500E+06	2.2500E+06	0.0	OK

BEAM MAX/MIN LOAD EFFECTS

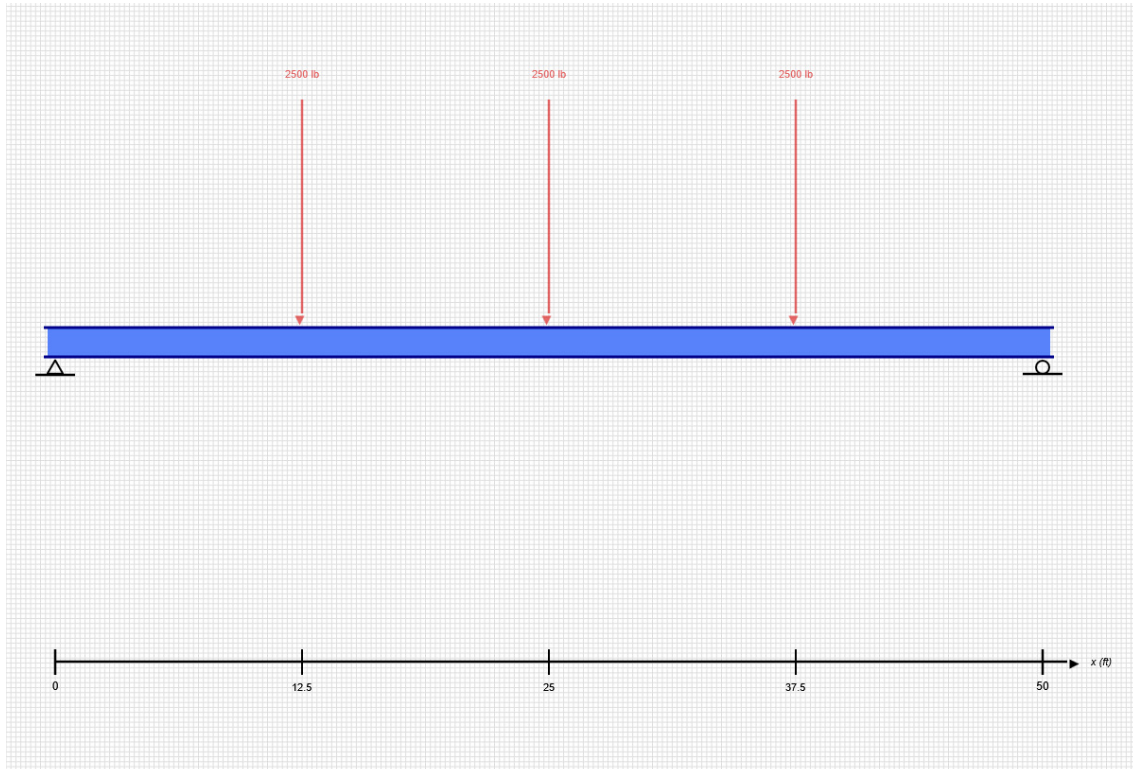
Item	Position (X)	Value
(---)	(L)	(F or F-L)
Max +V	0.00	3,750.0
Max -V	450.00	-3,750.0
Max +M	300.00	750,000.0
Max -M	0.00	0.0
Max Defl	300.00	-1.439

Value
(---)
3.75 kip
-3.75 kip
62.50 kip-ft
0.00 kip-ft
-1.439 in



SKYCIV BEAM ANALYSIS REPORT

Load Combination: DL



Software: SkyCiv Beam v3.3.2
Tue Jun 09 2026 12:52:28 GMT-0400 (Eastern Daylight Time)

Project Info

File Name: 01-Steel-Beam

Included in this Report:

- Input Summary
- Beam Section
- Free Body Diagram (FBD)
- Analysis Summary
- Analysis Results

SKYCIV FREE TRIAL EDITION

INPUT SUMMARY

General Info

Beam Length:	50 ft
Section Name:	W14x61
Self Weight:	False

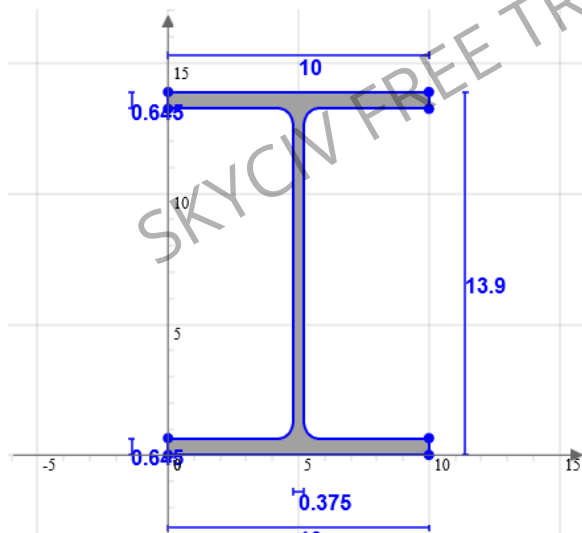
Supports

Support Type	Location
Pinned	0 ft
Roller	50 ft

Loads

Load Type	Location	Magnitude	Load Case
Point Load	12.5 ft	-2500 lb	DL
Point Load	37.5 ft	-2500 lb	DL
Point Load	25 ft	-2500 lb	DL

Beam Section



Geometric Properties		
A	17.9	in ²
C _z	5	in
C _y	6.95	in

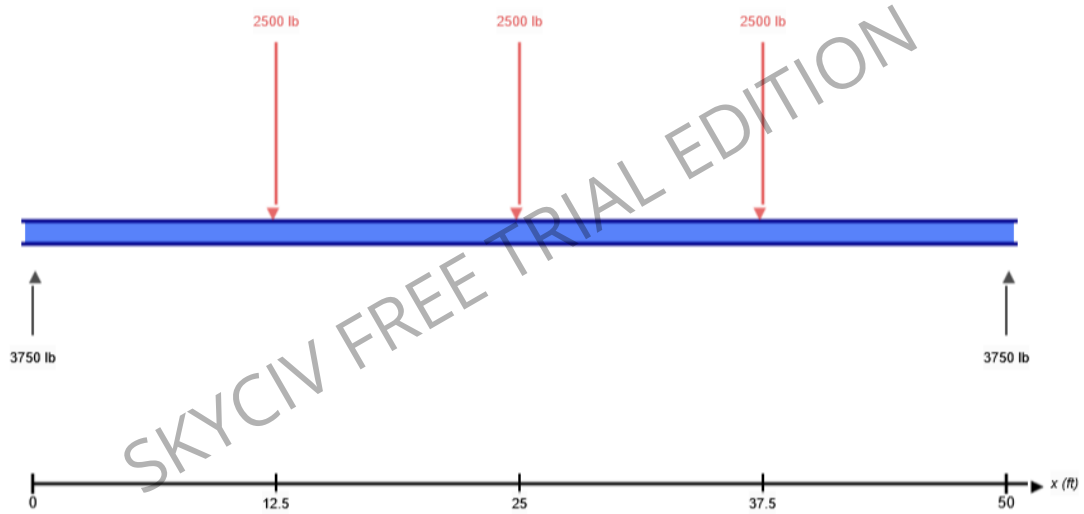
Bending Properties		
I _z	640	in ⁴
I _y	107	in ⁴

Shear Properties		
A _z	11.277	in ²
A _y	4.99	in ²

Torsion Properties		
J	2.19	in ⁴
r	0.991	in

Shape	Material	E (ksi)	v	ρ (lb/ft ³)
W14x61	Structural Steel	29000	0.27	490

FREE BODY DIAGRAM



RESULT SUMMARY

Check	Status	Limit	Ratio	Max
Deflection	PASS	L/250	0.601	L/416
Custom Stress Limit	PASS	39 ksi	0.209	8.145 ksi
Material Yield	PASS	36 ksi	0.226	8.145 ksi
Material Strength	PASS	58 ksi	0.14	8.145 ksi

ANALYSIS RESULTS

Reactions

Support at	R _x (lb)	R _y (lb)	M (kip-ft)
0	0	3750	0
50	0	3750	0

Force Extremes

Result	Max	Min
Bending Moment	62.5 kip-ft	0 kip-ft
Shear	3750 lb	-3750 lb
Displacement Y	0 in	-1.44 in

Stress Extremes

Result	Max (ksi)	Min (ksi)
Bending Stress	8.145	-8.145
Shear Stress Total	0.801	0.267
Max Combined Normal Stress	8.145	0
Min Combined Normal Stress	0	-8.145